



Computer Engineering Department
Computer Engineering
IC6400 - Operations Research

Project 0

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Demonstration Mode

Conditions

This file was generated to see the results of three different algorithms solving a random generated 0/1 Knapsack Problem. The problem has specific conditions everytime the `knapsack` program is run.

- The knapsack has a maximum size of 15.
- There are only 6 items.
- Each value for the items ranges between 0 to 20.
- Each cost for the items ranges between 0 to 7.

Algorithms

Dynamic Programming Algorithm

This algorithm uses a table to get to the optimal solution for the generated 0/1 Knapsack Problem. The strategy here is to solve subproblems, write down the optimal solution for that subproblem on a table and by using those solutions build the solution for the original problem. It always find the optimal solution for a 0/1 Knapsack Problem, as explained in course of Operations Research.

Basic Greedy Algorithm

This algorithm focuses on trying to put inside the knapsack the most valuable item, if it does not fit in, tries with the next most valueable item until the are no more items available.

Proportional Greedy Algorithm

It has the same strategy as the Basic Greedy, but it focuses on trying to put inside items based by their proportion between its value and cost. The item with highest proportion goes first, if it does not fit, goes to the next item with the highest proportion.

Demonstration Results

Problem generated

$$\begin{aligned} \text{Max } Z &= 20x_1 + 1x_2 + 5x_3 + 20x_4 + 13x_5 + 11x_6 \\ \text{such that : } &3x_1 + 2x_2 + 4x_3 + 4x_4 + 5x_5 + 5x_6 \leq 15 \\ &x_1, x_2, x_3, x_4, x_5, x_6 \in \{0, 1\} \end{aligned}$$

Dynamic Programming Solution

This table is generated using dynamic programming, based on the 0/1 Knapsack Problem algorithm covered in class.

Knapsack	Items					
	x_1	x_2	x_3	x_4	x_5	x_6
0	0	0	0	0	0	0
1	0	0	0	0	0	0
2	0	1	1	1	1	1
3	20	20	20	20	20	20
4	20	20	20	20	20	20
5	20	21	21	21	21	21
6	20	21	21	21	21	21
7	20	21	25	40	40	40
8	20	21	25	40	40	40
9	20	21	26	41	41	41
10	20	21	26	41	41	41
11	20	21	26	45	45	45
12	20	21	26	45	53	53
13	20	21	26	46	53	53
14	20	21	26	46	54	54
15	20	21	26	46	54	54

$$Z = 54, x_1 = 1, x_2 = 1, x_3 = 0, x_4 = 1, x_5 = 1, x_6 = 0$$

Greedy Algorithm

Basic Greedy Solution

This is the table for the Basic Greedy Algorithm, in which the most valuable item that fits into the remaining space in the knapsack is selected.

	Items					
Knapsack	x_1	x_4	x_5	x_6	x_3	x_2
Cost	3	4	5	5	4	2
Value	20	20	13	11	5	1

$$Z = 54, x_1 = 1, x_4 = 1, x_5 = 1, x_6 = 0, x_3 = 0, x_2 = 1$$

Proportional Greedy Solution

This table employs the Proportional Greedy Algorithm, in which the proportion of each item is calculated as its value divided by the capacity it consumes. Then, the item with the highest proportion that fits within the remaining knapsack capacity is selected at each step.

	Items					
Knapsack	x_1	x_4	x_5	x_6	x_3	x_2
Cost	3	4	5	5	4	2
Value	20	20	13	11	5	1
Proportion	6.666667	5.000000	2.600000	2.200000	1.250000	0.500000

$$Z = 54, x_1 = 1, x_4 = 1, x_5 = 1, x_6 = 0, x_3 = 0, x_2 = 1$$